STAT 215B Project Proposal

A Statistical Audit of LLM Calibration Heterogeneity Across Knowledge Domains

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Overview

LLM calibration is nearly always reported as a single aggregate scalar. In this project, we ask whether that aggregate masks systematic miscalibration in specific knowledge domains, and develop a statistically rigorous framework to detect and characterize that heterogeneity. The framing is analogous to bias auditing in the fairness literature: just as reporting a single accuracy number across demographic groups is insufficient for a fairness audit, reporting aggregate expected calibration error (ECE) is insufficient for a calibration audit. We apply multilevel regression, calibration curve clustering, empirical Bayes shrinkage, and multiple testing correction to the MMLU benchmark, using token-level log probabilities to elicit model confidence.

Research Questions

1. How much of the variance in LLM calibration error is attributable to domain vs. subject vs. individual question?
2. Which question- and subject-level features predict the direction and magnitude of miscalibration?
3. Are calibration patterns consistent across models, or model-specific?

Introduction and Background

Large language models (LLMs) are increasingly deployed in high-stakes settings (e.g. healthcare, law, education) where the reliability of model-expressed confidence is as important as accuracy itself. A well-calibrated model is one whose stated confidence tracks its empirical accuracy: if the model assigns 80% probability to an answer, it should be correct 80% of the time. Miscalibration, particularly overconfidence, means that users who rely on model confidence as a decision signal will be systematically misled.

A substantial literature has developed around LLM calibration. Guo et al. [2017] established the modern empirical study of neural network calibration, popularizing ECE as a standard metric and demonstrating the surprising effectiveness of temperature scaling as a post-hoc correction. Kadavath et al. [2022] showed that for multiple-choice tasks, token-level log probabilities provide a meaningful calibration signal, and that base (pre-RLHF) language models exhibit better-calibrated token probabilities than their RLHF-finetuned counterparts—though the gap can be partially remediated by temperature adjustment. Subsequent work has studied the impact of instruction tuning on calibration [Nakkiran et al., 2025, Yaldiz et al., 2026]. Xiong et al. [2024] provided a systematic empirical benchmark of confidence elicitation methods across task types, finding that all methods "struggle in challenging tasks, such as those requiring professional knowledge"—a finding that suggests calibration quality may vary by knowledge domain, though the authors do not investigate this directly.

Much of the subsequent literature has focused on two directions: characterizing average miscalibration across models and elicitation strategies, and developing post-hoc methods to correct it. Recent recalibration work has somewhat acknowledged domain-level heterogeneity: Luo et al. [2025] explicitly fits a subject-specific temperature parameter for each of MMLU's 57 subjects, implicitly recognizing that a single global correction is insufficient. Tan et al. [2026] takes a complementary approach, using base model signals to recalibrate instruction-tuned models globally, and evaluates on MMLU as a benchmark.

What is missing is a statistical audit of that heterogeneity. To our knowledge, the existing LLM calibration literature reports subject-level ECE without attempting to (i) stabilize those estimates against large differences in per-subject sample size, (ii) formally test which subjects are robustly miscalibrated after correcting for multiple comparisons, (iii) decompose how much of the total variance in calibration error is attributable to domain versus subject versus individual question, or (iv) characterize subjects by their *pattern* of miscalibration rather than its scalar magnitude. This is analogous to the gap between reporting per-group accuracy in a fairness audit—which the fairness literature has long recognized as insufficient [Barocas et al., 2023]—and conducting a rigorous audit with appropriate uncertainty quantification.

This project imports the logic of subgroup auditing into calibration evaluation. The answer matters for downstream use: if miscalibration is diffuse, a single global recalibration (e.g., temperature scaling) is a reasonable fix. If it is concentrated in specific domains—professional law, abstract mathematics, clinical medicine—then users relying on model confidence in those domains are exposed to systematically higher risk, and domain-aware recalibration methods would be warranted. More broadly, this work contributes to the question of what it means to audit an AI system's reliability rather than merely benchmark its average performance.

Data

MMLU [Hendrycks et al., 2021] (`cais/mmlu` on HuggingFace): ∼14,000 four-option multiple-choice questions across 57 subjects nested in 4 broad domains (STEM, Social Sciences, Humanities, Other). The three-level hierarchy—questions → subjects → domains—is natural for the proposed analysis.

Models: GPT-4o-mini via the OpenAI API (∼\$0.50 for the full dataset) and Llama-3-8B via HuggingFace (free). Running both allows us to distinguish task-driven from model-specific miscalibration.

Methods

The following describes one possible methodology for addressing our research questions. We may decide to revise our modeling choices in the future prior to beginning our analysis.

Step 1: Confidence Elicitation

For each question, we prompt the model to respond with a single letter (A/B/C/D) and extract the token-level log probabilities over the four answer tokens. Applying softmax yields a proper probability distribution over options; the model's confidence $c_i \in [0, 1]$ is the maximum of this distribution. This is the standard approach in calibration research [Kadavath et al., 2022] and avoids the noise of verbal elicitation. Critically, repeated querying is unnecessary: the token probability is the model's stated distribution, not an empirical frequency to be estimated by sampling.

Step 2: Outcome and Features

The primary outcome is the *signed calibration gap* at the question level: $\text{gap}_i = c_i - y_i$, where $y_i \in \{0, 1\}$ is correctness. This is positive under overconfidence and negative under underconfidence, and enables regression-based inference that binning-based ECE cannot support. Alternatively, the analysis could be replicated using question-level calibration scores such as log-likelihoods, as ECE is not a proper scoring rule. The choice will be finalized during analysis; the modeling framework in subsequent steps applies to any question-level calibration outcome.

Question-level covariates, for example: word count, average and max option length, negation indicator (`not`, `except`, `least`, `never`), entropy of the probability distribution.

Subject-level covariate: empirical model accuracy within subject (difficulty proxy).

Step 3: Nonparametric Calibration Curves

For each subject, we estimate a reliability curve using isotonic regression—a nonparametric monotone fit of accuracy on confidence—rather than arbitrary equal-width binning. The monotonicity constraint is theoretically motivated and acts as a regularizer, but is an assumption rather than a guarantee. We therefore fit an unconstrained kernel smoother alongside each isotonic curve; substantive divergences between the two identify subjects where the monotonicity assumption may distort the estimated curve, and are themselves interpretable findings. A scalar calibration summary per subject is derived from the fitted curve; the specific measure will be consistent with the scoring rule chosen in Step 2. (ECE per subject is computed as the integrated absolute deviation between the isotonic fit and the diagonal.)

Step 4: Multilevel Regression, Shrinkage, and Multiple Testing

We model the question-level outcome directly across all ∼14,000 questions using a three-level mixed model with questions nested in subjects nested in domains:

$$\text{outcome}_{ijk} = \mu + \mathbf{x}_{ijk}^\top \boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk},$$

where $\mathbf{x}_{ijk}$ are question-level covariates, $u_k \sim \mathcal{N}(0, \sigma^2_{\text{domain}})$ is a domain random effect, $v_{jk} \sim \mathcal{N}(0, \sigma^2_{\text{subject}})$ is a subject-within-domain random effect, and ε_{ijk} is residual noise. The fixed effects $\boldsymbol{\beta}$ characterize which question features are associated with miscalibration across the full dataset. The variance components $(\sigma^2_{\text{domain}}, \sigma^2_{\text{subject}}, \sigma^2_{\varepsilon})$ answer RQ1 via an ICC-style decomposition—the fraction of total calibration heterogeneity attributable to each level of the hierarchy. We interpret $\boldsymbol{\beta}$ as descriptive associations rather than causal effects.

Subject-level shrunken calibration estimates are derived directly from the fitted random effects $\hat{v}_{jk}$, which implicitly borrow strength across subjects in proportion to sample size—serving the same purpose as a separate empirical Bayes shrinkage step. We then test each of the 57 subjects for significant miscalibration using these estimates and apply the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] to control the false discovery rate, yielding a ranked list of subjects where miscalibration is statistically established rather than just descriptively apparent.

Step 5: Cross-Model Replication

The full pipeline is run on both GPT-4o-mini and Llama-3-8B (base). Because the base Llama model does not follow instructions, it will be prompted using a completion-style format rather than an instruction prompt; this difference in elicitation format will be noted as a limitation when interpreting cross-model comparisons. Subjects flagged as miscalibrated by both models suggest task-driven miscalibration reflecting question structure; subjects that diverge suggest model-specific effects of training. We compare subject-level calibration rankings across models using rank correlation with bootstrap confidence intervals.

Deliverables

1. **Variance decomposition**: Fraction of calibration heterogeneity attributable to domain, subject, and question levels via ICC from the multilevel model.
2. **FDR-corrected subject rankings**: Subjects robustly identified as miscalibrated after shrinkage and multiple testing correction.
3. **Predictors of miscalibration**: Question-level features associated with over- vs. underconfidence.
4. **Cross-model agreement**: Whether miscalibration patterns are task-driven or model-specific.

Feasibility

Component	**Time**	**Cost**
Download MMLU + feature extraction	1–2 hrs	Free
API calls (14k questions, two models)	~2 hrs runtime	~$0.50
Isotonic + kernel calibration curves	2–3 hrs	—
Multilevel regression	2–3 hrs	—
EB shrinkage + multiple testing	2 hrs	—
Cross-model comparison + writing	—	—

The entire pipeline runs on a laptop with no GPU required. Data collection can run overnight. All methods are implementable in R (`lme4`, `fdrtool`) or Python (`sklearn`, `scipy`, `statsmodels`).

We will have one person lead the statistical modeling components — the multilevel regression, empirical Bayes shrinkage, and multiple testing — while the other will lead data collection, API querying, and the calibration curve estimation pipeline. Writing will be shared. We expect roles to overlap and will adjust on an ongoing, iterative basis as the shape of the project becomes more clear.

References

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